Maharashtra Education Society's
Institute of Management and Career Courses (IMCC), Pune
AUTONOMOUS

Academic Year 2026-27
FY MCA- SEM I
IT12 Data Structure and Algorithms
Practical Assignment1

Date: 15/10/2026 **Marks: 40**
Roll No: 2601182 **Name: Shekhar Mohan Sonawane**

Create a Singly Linear Linked List with following operations

1. Create Linked List (append)
2. Traverse and print the node values
3. Insert node at a specific position
4. Delete alternate nodes in the linked list

1.

```python
# 1. Create Linked List (Append)

class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

class LinkedList:
    def __init__(self):
        self.head = None

    def append(self, data):
        new = Node(data)

        if self.head is None:
            self.head = new
            return

        temp = self.head
        while temp.next:
            temp = temp.next

        temp.next = new

    def display(self):
        temp = self.head
        while temp:
            print(temp.data, end=" -> ")
            temp = temp.next
        print("None")


ll = LinkedList()
ll.append(10)
ll.append(20)
ll.append(30)

ll.display()
```

(variable) new: Node

Output:

```
10 -> 20 -> 30 -> None
```

2.

```
# 2. Traverse and Print Node Values

    def display(self):
        temp = self.head
        while temp:
            print(temp.data, end=" -> ")
            temp = temp.next
        print("None")


ll = LinkedList()
ll.append(10)
ll.append(20)
ll.append(30)

ll.display()
```

Output:

```
10 -> 20 -> 30 -> None
```

3.

```
# 3. Insert Node at a Specific Position

    def insert(self, data, pos):
        new = Node(data)

        if pos == 0:
            new.next = self.head
            self.head = new
            return

        temp = self.head
        for i in range(pos - 1):
            temp = temp.next

        new.next = temp.next
        temp.next = new


ll = LinkedList()
ll.append(10)
ll.append(20)
ll.append(30)

ll.insert(15, 1)
ll.display()
```

Output:

```
10 -> 15 -> 20 -> 30 -> None
```

4.

```
# 4. Delete Alternate Nodes

    def deleteAlternate(self):
        temp = self.head
        while temp and temp.next:
            temp.next = temp.next.next
            temp = temp.next


ll = LinkedList()
ll.append(10)
ll.append(20)
ll.append(30)
ll.append(40)
ll.append(50)

print("Before deleting:")
ll.display()

ll.deleteAlternate()

print("After deleting alternate nodes:")
ll.display()
```

Output:

```
Before deleting:
10 -> 20 -> 30 -> 40 -> 50 -> None
After deleting alternate nodes:
10 -> 30 -> 50 -> None
```